

sentiment analysis have shown the effectiveness of simple long-short strategies based on sentiment scores (Lopez-Lira and Tang, 2023). Further research on fine-tuned LLMs like FinGPT and OPT demonstrates improved performance through domain-specific alignment (Kirtac and Germano, 2024; Zhang et al., 2024a). Advanced methods involve summarizing news data and reasoning about their relationship with stock prices (Fatouros et al., 2024a; Wang et al., 2024b).

Reasoning-Driven Agents

Reasoning-driven agents enhance trading decisions through mechanisms like reflection and debate. Reflection-driven agents, such as FinMem (Yu et al., 2023) and FinAgent (Zhang et al., 2024b), use layered memorization and multimodal data to summarize inputs into memories, inform decisions, and incorporate technical indicators, achieving superior backtest performance while mitigating hallucinations (Ji et al., 2023). Debate-driven agents, like those in heterogeneous frameworks (Xing, 2024) and TradingGPT (Li et al., 2023b), enhance reasoning and factual validity by employing LLM debates among agents with different roles, improving sentiment classification and increasing robustness in trading decisions.

Reinforcement Learning-Driven Agents

Reinforcement learning methods align LLM outputs with expected behaviors, using backtesting as rewards. SEP (Koa et al., 2024) employs RL with memorization and reflection to refine LLM predictions based on market history. Classical RL methods are also used in trading frameworks that integrate LLM-generated embeddings with stock features, trained via algorithms like Proximal Policy Optimization (PPO) (Ding et al., 2023; Schulman et al., 2017).

2.3. LLMs as Alpha Miners

LLMs are also used to generate alpha factors instead of making direct trading decisions. QuantAgent (Wang et al., 2023) demonstrates this by leveraging LLMs to produce alpha factors through an inner-loop and outer-loop architecture. In the inner loop, a writer agent generates a script from a trader’s idea, while a judge agent provides feedback. In the outer loop, the code is tested in the real market, and trading results enhance the judge agent. This approach enables progressive approximation of optimal behavior.

Subsequent research, such as AlphaGPT (Wang et al., 2023), proposes a human-in-the-loop framework for alpha mining with a similar architecture. Both studies showcase the effectiveness of LLM-powered alpha mining systems, highlighting their potential in automating and accelerating the development of trading strategies by generating and refining alpha factors.

3. TradingAgents: Role Specialization

Assigning LLM agents clear, well-defined roles with specific goals enables the breakdown of complex objectives into smaller, manageable subtasks. Financial trading is a prime example of such complexity, demanding the integration of diverse signals, inputs, and specialized expertise. In the real world, this approach to managing complexity is demonstrated by trading firms that rely on expert teams to collaborate and make high-stakes decisions, underscoring the multifaceted nature of the task.

In a typical trading firm, vast amounts of data are collected, including financial metrics, price movements, trading volumes, historical performance, economic indicators, and news sentiment. This data is then analyzed by quantitative experts (quants), including mathematicians, data scientists, and engineers, using advanced tools and algorithms to identify trends and predict market movements.

Inspired by this organizational structure, TradingAgents defines seven distinct agent roles within a simulated trading firm: Fundamentals Analyst, Sentiment Analyst, News Analyst, Technical Analyst, Researcher, Trader, and Risk Manager. Each agent is assigned a specific name, role, goal, and set of constraints, alongside predefined context, skills, and tools tailored to their function. For example, a Sentiment Analyst is equipped with tools like web search engines, Reddit search APIs, X/Twitter search tools, and sentiment score calculation algorithms, while a Technical Analyst can execute code, calculate technical indicators, and analyze trading patterns. More specifically, TradingAgents assumes the following teams.

3.1. Analyst Team

The Analyst Team (Figure 2) is composed of specialized agents responsible for gathering and analyzing various types of market data to inform trading decisions. Each agent focuses on a specific aspect of market analysis, bringing together a comprehensive view of the market's conditions.

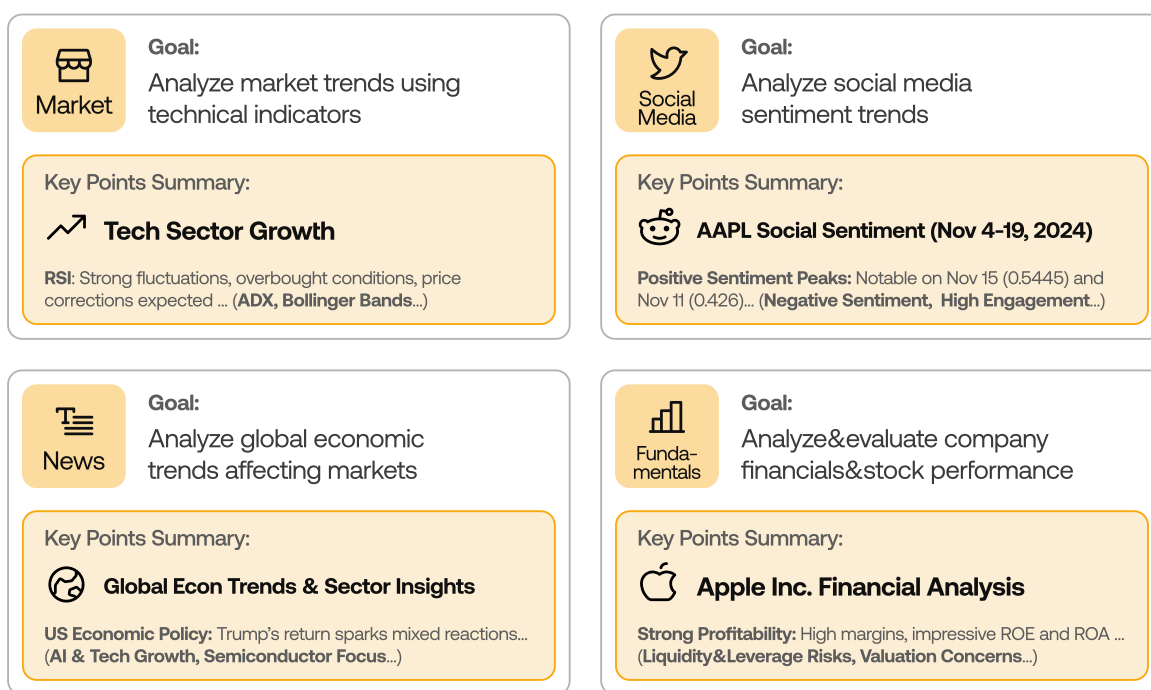


Fig. 2: TradingAgents Analyst Team

- **Fundamental Analyst Agents:** These agents evaluate company fundamentals by analyzing financial statements, earnings reports, insider transactions, and other pertinent data. They assess a company's intrinsic value to identify undervalued or overvalued stocks, providing insights into long-term investment potential.
- **Sentiment Analyst Agents:** These agents process large volumes of social media posts, sentiment scores, and insider sentiments derived from public information and social media activity. They gauge market sentiment to predict how collective investor behavior might impact stock prices in the short term.

- **News Analyst Agents:** These agents analyze news articles, government announcements, and other macroeconomic indicators to assess the market’s macroeconomic state, major world events, and significant company changes. They identify news events that could influence market movements, helping to anticipate sudden shifts in market dynamics.
- **Technical Analyst Agents:** These agents calculate and select relevant technical indicators, such as Moving Average Convergence Divergence (MACD) and Relative Strength Index (RSI), customized for specific assets. They analyze price patterns and trading volumes to forecast future price movements, assisting in timing entry and exit points.

Collectively, the Analyst Team synthesizes data from multiple sources to provide a holistic market analysis. Their combined insights form the foundational input for the Researcher Team, ensuring that all facets of the market are considered in subsequent decision-making processes.

3.2. Researcher Team

The Researcher Team (Figure 3) is responsible for critically evaluating the information provided by the Analyst Team. Comprised of agents adopting both bullish and bearish perspectives, they engage in multiple rounds of debate to assess the potential risks and benefits of investment decisions.

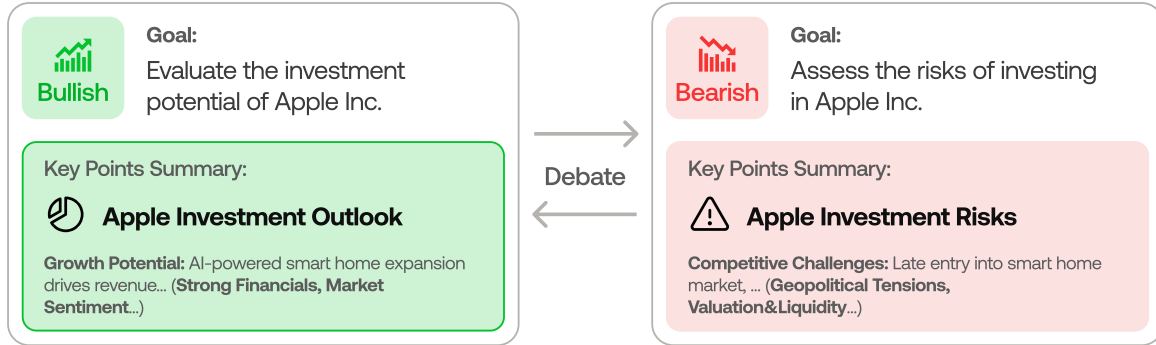


Fig. 3: TradingAgents Researcher Team: Bullish Perspectives and Bearish Perspectives

- **Bullish Researchers:** These agents advocate for investment opportunities by highlighting positive indicators, growth potential, and favorable market conditions. They construct arguments supporting the initiation or continuation of positions in certain assets.
- **Bearish Researchers:** Conversely, these agents focus on potential downsides, risks, and unfavorable market signals. They provide cautionary insights, questioning the viability of investment strategies and highlighting possible negative outcomes.

Through this dialectical process, the Researcher Team aims to reach a balanced understanding of the market situation. Their thorough analysis helps in identifying the most promising investment strategies while anticipating possible challenges, thus aiding the Trader Agents in making informed decisions.

3.3. Trader Agents

Trader Agents (Figure 4) are responsible for executing trading decisions based on the comprehensive analysis provided by the Analyst Team and the nuanced perspectives from the Researcher Team. They